

AMPKG: Extracting Concept Prerequisites from Multimodal Educational Media

Anonymous Submission to IEEE DSAA 2026

Abstract—Recovering the prerequisite structure of a subject automatically from raw lecture video is hard. Transcript-only methods miss what the lecturer draws rather than says, and large language models tend to invent concepts and to point prerequisite arrows in the wrong direction. We present AMPKG, a framework that builds a concept-prerequisite graph from a video, its transcript, and any slides in three phases: it first aligns all three sources into a single timestamped sequence, then routes each segment to either a cheap local model or a language model based on how ambiguous it is, and finally extracts concepts and validates each prerequisite edge against four independent pieces of evidence before adding it. Two design choices do most of the work. Every concept kept in the graph must point back to a specific timestamped moment in the source, which removes invented concepts by construction. And each prerequisite edge must agree with temporal, visual, discourse, and evidence-strength signals, which fixes the direction the language model gets wrong. On a new benchmark of 120 annotated Khan Academy lectures across four STEM subjects, this raises directed edge F_1 from 0.031 for the best language-model baseline to 0.512, while keeping every surviving concept traceable to the lecture it came from.

Index Terms—Pedagogical knowledge graphs, prerequisite inference, multimodal learning, evidence grounding, educational AI

I. INTRODUCTION

A lecture is more than the words the instructor says. Over fifteen minutes the lecturer speaks, writes an equation on the board one line at a time, and points to a slide, and a student who has followed it knows not only the facts about light but the order in which they depend on one another: refractive index comes before Snell’s law, and Snell’s law comes before the critical angle. We want to recover that dependency structure automatically. Platforms such as Khan Academy and MIT OpenCourseWare host hundreds of thousands of hours of lectures, and tutoring systems, curriculum planners, and retrieval pipelines all need to know which concepts a lecture teaches and which of them must be learned first. A concept-prerequisite knowledge graph [3], [20] is the natural way to record this, and prerequisite relations among educational concepts have been studied for MOOCs and academic text [17], [18], [19] and used to drive knowledge tracing [21], [22]. Building one from raw video, however, is harder than it looks, and current methods fail in three recurring ways.

The first failure is missing content. Transcript-only extraction, still the common practice and typically built on open relation extraction or LLM prompting [36], [9], [4], captures nothing the lecturer draws instead of says, even though derivations and diagrams carry much of the teaching

in a STEM lecture. Multimodal knowledge graphs that fuse images or video with text [1], [2], [5], [6] address part of this gap, but not for the time-ordered structure of a lecture. Methods that do read the video usually sample frames on a fixed clock, so they miss a board that is rewritten faster than the sampling rate and waste effort on long stretches where nothing changes.

The second failure is invented concepts. Given an optics transcript, a language model will readily list “total internal reflection” or “Huygens’ principle” as key concepts even when the lecture never mentioned them, because it is drawing on what it knows about optics rather than on what was taught. This hallucination is a known hazard when language models build knowledge graphs [8], [7], [10], [11]. In most settings it is a nuisance; in education it is dangerous, because a tutor that asserts a wrong learning dependency misleads students in a way that still looks reasonable.

The third failure is the direction of the edge. Even when both concepts are identified correctly, deciding which one is the prerequisite is surprisingly unreliable. The obvious rule, that the concept mentioned earlier is the prerequisite, breaks often: lecturers preview material before defining it, return to foundational ideas to reinforce them, and motivate a concept with an example before stating it. When we replace our edge-validation step with a direct language-model judgment, directed edge F_1 falls by 63.5%. This is not a question of model size. The three language models we tested, from 4B to 9B parameters, all score about the same, so the failure is structural rather than a matter of capacity.

AMPKG addresses all three failures by not asking one model to do everything. Each step has a defined input, a defined output, and a way to check whether its output can be trusted. A single timestamped representation, the Evidence Unit, aligns transcript text, video frames, and slide regions, so later steps reason over one sequence rather than three formats. A routing step spends expensive language-model inference only on segments a cheap embedding model finds ambiguous. And Evidence-Weighted Prerequisite Inference (EWPI) checks each candidate edge against four independent signals before admitting it. The paper makes three contributions.

- 1) Evidence-Weighted Prerequisite Inference, a voting rule that combines temporal order, visual board progression, discourse cues, and an evidence-strength ratio to decide both whether an edge is real and which way it points. It raises directed edge F_1 from 0.031 for the best language-model baseline to 0.512, and we show no single signal

accounts for the gain.

- 2) Provenance enforcement, an architectural rule that every concept kept in the graph must link to at least one timestamped Evidence Unit. Concepts that cannot be deleted, which removes invented concepts by construction and leaves the surviving graph fully traceable to its source.
- 3) Ambiguity-band routing, which classifies each segment by how clearly it maps to a subject and sends only the unclear ones to the language model. On a 40-lecture course this cuts language-model calls from every segment to roughly 45% with no measurable loss in quality.

We also release the benchmark: 120 Khan Academy lectures across four STEM subjects, with 1,847 annotated concept-prerequisite pairs and inter-annotator agreement $\kappa = 0.74$, the largest expert-annotated prerequisite benchmark for multimodal lecture video that we are aware of. Section II describes the three phases; Section III reports results, ablations, and significance tests.

II. METHODOLOGY

A. System Overview

AMPKG takes a lecture video, its transcript, and any slide deck and produces a concept-prerequisite graph in three phases (Fig. 1). The phases are not just an organizing convenience: each solves a different problem, and the output of one is required input for the next.

Phase A handles the fact that the three inputs are in different formats and carry information at different rates. It normalizes them into one representation, the Evidence Unit: a short timestamped window carrying the aligned transcript text, a video frame or slide region, a one-line description of what is new in that visual, and a dense embedding. Every later step reasons over this single sequence rather than over the raw inputs.

Phase B handles cost. Sending every segment of an hour-long lecture to a language model is expensive and, as our ablation shows, mostly wasted, because many segments are routine restatements or transitions that a small embedding model labels correctly on its own. Phase B measures how ambiguous each segment is and escalates only the unclear ones.

Phase C builds the graph and is where most of the difficulty lies. It extracts concepts, merges the many surface forms of the same concept into one node, proposes candidate prerequisite edges, and then validates each candidate with EWPI before admitting it. A final provenance step deletes any concept that the language model named but that cannot be traced to an actual moment in the lecture. Table I lists the symbols used throughout.

B. Phase A: Building Evidence Units

The Evidence Unit is the data structure everything else is built on. Each unit is a six-tuple

$$u_i = (\tau_i, x_i, f_i, c_i, \Delta_i, m_i), \quad (1)$$

TABLE I
NOTATION USED THROUGHOUT THE PAPER.

Symbol	Meaning
u_i	Evidence Unit: one timestamped multimodal window, Eq. (1)
τ_i	modality of u_i : video frame, transcript span, or slide region
x_i	transcript text aligned to u_i
f_i	visual of u_i : frame image or slide region
c_i	caption: what is new in f_i
Δ_i	time interval $[t_{\text{start}}, t_{\text{end}}]$ of u_i
m_i	metadata of u_i : source file, slide/page number
$\mathcal{U}$	full sequence of Evidence Units for a lecture
$\mathcal{U}_A$	Evidence Units that support concept A
e_{AB}	candidate edge “ A requires B ” ($A \rightarrow B$)
T, V, D, E	the temporal, visual, discourse, and evidence-strength signals
$w_T, \dots, w_E$	weights on the four signals
λ	minimum EWPI score to admit an edge
ε	minimum margin over the reverse edge
γ	routing margin: clear vs. ambiguous segment

whose fields are: the modality τ_i (a video frame, a transcript span, or a slide region); the aligned transcript text x_i for that window; the visual f_i (the frame image or slide region); the caption c_i , a short description of what is new in f_i ; the time interval $\Delta_i = [t_{\text{start}}, t_{\text{end}}]$; and the metadata m_i , which records the source file and, for slides, the slide and page number. The full sequence $\mathcal{U} = \{u_1, \dots, u_N\}$ is consumed by every later phase, and the rest of this subsection explains how each field is filled.

The visual f_i comes from frame selection, which is the part that fixed-interval sampling gets wrong, and it gets it wrong in both directions. A lecturer may spend forty-five seconds writing one derivation step, so a frame every second yields thirty near-identical images; conversely a board can be wiped and rewritten in under two seconds, which a sparse sampler misses entirely. We instead select frames by content change. Raw frames are taken at 1 Hz for slide lectures and 2 Hz for chalkboard sessions, then reduced by three filters. Dark, blurry, and pure talking-head frames are dropped first, using simple intensity and structural-similarity checks [35] with no neural cost. Remaining frames are deduplicated by colour-histogram correlation against the last sixty seconds; anything above 0.98 is treated as a repeat. What survives is embedded with CLIP [23] and clustered per video, and the frame nearest each cluster centre is kept, with a preference for text-dense frames so that selection favours frames that teach something rather than merely typical ones.

The transcript text x_i and the interval Δ_i come from speech recognition with Qwen3-ASR-0.6B, in the lineage of large-scale weakly supervised recognition [38], which gives word-level timestamps; an existing subtitle file is used directly when present. Those timestamps set Δ_i and let each text span be aligned to the frame on screen at the same moment, and the ordering of the Δ_i across units is what the temporal signal in EWPI later reads. The caption c_i comes from a vision-language model [24], [25], [26], here Gemma-4-E4B-

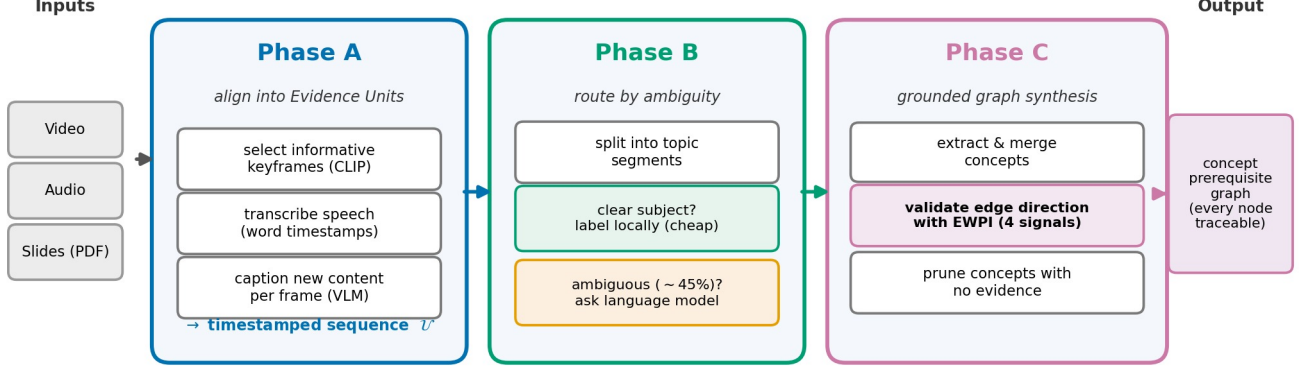


Fig. 1. The three phases of AMPKG. *Phase A* (left) converts the video, audio, and slides into a single sequence of timestamped Evidence Units $\mathcal{U}$, selecting informative keyframes, transcribing speech, and captioning each frame with only the content that is new relative to the previous one. *Phase B* (lower left) groups the units into topic segments and scores each against the candidate subjects; a segment that maps clearly to one subject is labelled locally, and only an ambiguous segment (about 45% of them) is sent to the language model. *Phase C* (right) extracts concepts and relations, merges duplicate concept names, proposes prerequisite edges, and admits an edge only if EWPI confirms its direction; provenance enforcement then drops any concept with no supporting evidence.

it: rather than describe each frame on its own, we pass the previous kept frame as context and ask only for content that has newly appeared, which avoids repetitive captions such as “the lecturer is writing on the board.” Each unit also stores a dense embedding of x_i and c_i [27], [28], [29] that serves as the anchor a concept points back to, while m_i lets a returned concept be cited down to the exact slide or video file it came from.

C. Phase B: Ambiguity-Band Routing

Routing needs the lecture broken into segments, each covering one topic or teaching move. We merge consecutive units with a sentence-window parser and place a boundary wherever any of four signals fires: a drop in cosine similarity between adjacent unit embeddings, a changepoint found by PELT on the leading PCA components of the embedding sequence, a switch in source (read from the metadata m_i , for instance moving from video to a new slide deck), or a hard cap on segment length. The union is deliberately generous, because an oversplit segment is easy to recover from later, whereas a segment that fuses two topics is not.

For each segment we measure how clearly it belongs to one subject. The segment’s embedding $\phi(u_i)$ is compared to a centroid $\mathbf{v}_d$ for each candidate subject d , $s_i^{(d)} = \cos(\phi(u_i), \mathbf{v}_d)$, and we take the gap g_i between the top two scores as a measure of certainty. A wide gap means the segment plainly belongs to one subject; a narrow gap means it is ambiguous. The routing rule is

$$R(g_i) = \begin{cases} \text{accept top subject locally,} & g_i > \gamma \\ \text{ask the language model,} & g_i \leq \gamma \end{cases} \quad (2)$$

with $\gamma = 0.25$ set from the training-split score distribution. An escalated segment goes to Gemma-4-E4B-it [39], served with

PagedAttention [37], which returns the subject, a teaching role (such as core topic, prerequisite revision, or transition), a keep-or-drop decision, and a relevance score. A short smoothing pass then rescues isolated mistakes: a borderline-dropped segment surrounded by kept ones is restored, since a real topic shift rarely shows up as a single off-topic segment between on-topic neighbours. In practice this sends about 45% of segments to the language model instead of all of them, which on a 40-lecture course is roughly a 55% cut in inference cost; Section III-C confirms it costs nothing in quality.

D. Phase C: Grounded Graph Synthesis

We query the language model three times per segment rather than once, because asking for concepts, relations, and prerequisites together produced inconsistent output: the model would name a concept in the relation step that it had not listed as a concept, inventing nodes midway. Separating the queries and feeding each result to the next as a fixed vocabulary largely stops this. The first query lists only concepts stated explicitly in the segment, each typed as a definition, law, formula, procedure, example, or fact, with a confidence in $[0, 1]$. The second query, given that concept list, extracts typed relations among those concepts (such as *explains* or *formula_for*). The third, given concepts and relations, proposes prerequisite pairs “A requires B.” These pairs are only candidates; none enters the graph until EWPI validates it.

The same concept appears under many names even in one lecture: “angle of incidence,” “incidence angle,” “ θ_i ,” or just “the angle.” Left alone, each becomes a separate disconnected node. This is the entity-resolution problem studied at length for structured data [12], [13], [14], [15], [16]; our setting is lighter, so we resolve names in three steps over all extracted concepts at once. Names are normalized and a small synonym dictionary is applied (so “law of refraction” meets “Snell’s

law”); pairs whose fuzzy token ratio is at least 88 are merged automatically, and the rest are clustered [34] before adjudication. Pairs scoring between 70 and 88 are genuinely uncertain, so the language model decides, given both names and their definitions, whether to merge them. Each resulting cluster then elects a canonical name by

$$\text{Score}(c) = 0.6 \text{conf}(c) + 0.4 \widetilde{\text{freq}}(c), \quad (3)$$

where $\text{conf}(c)$ is the extraction confidence and $\widetilde{\text{freq}}(c)$ is occurrence frequency normalized to its maximum. The canonical node absorbs the aliases and collects the supporting Evidence Units from every modality in which the concept appeared.

This last point matters for what follows. Because aliases are merged, Snell’s law can be supported at once by a spoken transcript span, a board-written formula in a frame, and a slide caption, and all three links attach to one node. The count of distinct supporting units, and how many modalities they span, is exactly the evidence-strength signal E that EWPI uses next: a concept grounded in three modalities is better attested than one mentioned in passing.

E. Evidence-Weighted Prerequisite Inference (EWPI)

The candidate edges from the third query cannot be trusted as they stand: as Section I noted and Section III-D quantifies, a language model points the prerequisite arrow the wrong way about 75% of the time when it has only the transcript to go on. EWPI instead decides each edge from four signals that look at the lecture in four different ways, so that no single misleading cue (such as mention order) can carry an edge on its own. For a candidate edge e_{AB} , read as “ A requires B ,” each signal lies in $[0, 1]$ and is illustrated in Fig. 2.

Temporal order T is the fraction of the lecture in which B first appears before A ; a value near 1 means B consistently comes first, which supports $A \rightarrow B$. *Visual progression* V is the fraction of kept frames in which B ’s term or formula is on the board before A is first written, and is set to zero when neither concept is written on a board. *Discourse* D is the fraction of transcript spans in which B sits in a dependency or coreference chain that precedes A , read off the spaCy parse of the aligned text. *Evidence strength* E compares how well each concept is attested,

$$E(e_{AB}) = \min\left(1, \frac{|U_B|}{|U_A|}\right), \quad (4)$$

on the observation that a prerequisite is usually grounded in more units and more modalities than the concept that builds on it.

The four signals combine linearly,

$$\text{EWPI}(e_{AB}) = w_T T + w_V V + w_D D + w_E E, \quad (5)$$

with weights $(w_T, w_V, w_D, w_E) = (0.35, 0.25, 0.25, 0.15)$ from a grid search on the training split. The edge is admitted only when it is both well supported in absolute terms and clearly better than its reverse,

$$\text{EWPI}(e_{AB}) \geq \lambda \quad \text{and} \quad \text{EWPI}(e_{AB}) - \text{EWPI}(e_{BA}) \geq \varepsilon, \quad (6)$$

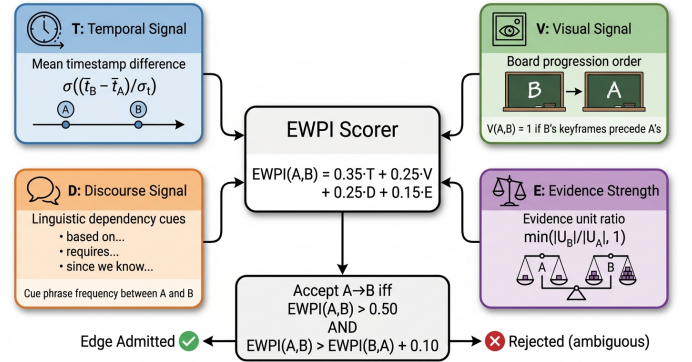


Fig. 2. How EWPI decides the edge “ A requires B .” Four signals vote: temporal order (T , does B tend to be mentioned before A), visual board progression (V , is B written on the board before A), discourse cues (D , phrases such as “based on” or “requires” linking the two), and evidence strength (E , the ratio of supporting Evidence Units). Their weighted sum is the EWPI score. The edge is admitted only if that score passes 0.50 and beats the reverse edge by at least 0.10; otherwise the pair is left unconnected, which is what prevents the language model’s wrong-direction guesses from entering the graph.

TABLE II
HYPERPARAMETER VALUES, ALL SET ON THE 84-VIDEO TRAINING SPLIT AND NEVER ON THE TEST SET.

Parameter	Value	Set by
λ (edge acceptance)	0.50	grid search
ε (directional margin)	0.10	grid search
signal weights	see Eq. (5)	grid search
γ (routing margin)	0.25	score distribution
segment boundary threshold	0.35	validation set
name-merge threshold	≥ 88	validation set
name-merge gray band	[70, 88)	validation set
provenance similarity	0.80	embedding calibration

with $\lambda = 0.50$ and a directional margin $\varepsilon = 0.10$ (Table II lists these and the other thresholds). If neither direction clears both bars, the pair is left unconnected rather than guessed. On our benchmark this margin rule flips the direction of 74% of the raw candidate edges, in line with the error rate of Section III-D.

F. Bipartite Provenance Enforcement and DAG Construction

Even after EWPI, some concept nodes carry no supporting Evidence Unit. These are the concepts the language model named while inferring prerequisites but that never actually appeared in a transcript span, a frame caption, or a slide; it produced them from what it knows about the subject, not from the lecture. The provenance rule deletes any such node, together with its edges, in a single pass over the bipartite graph of concepts and Evidence Units. Figure 3 shows the case from the introduction: refractive index, Snell’s law, and total internal reflection each link to evidence and stay, while “Huygens’ principle,” which the lecture never covered, has no link and is pruned. The choice favours precision over recall, on the view that a small trustworthy graph is more useful than

Bipartite Provenance Enforcement in Multi-Modal Knowledge Graph Extraction

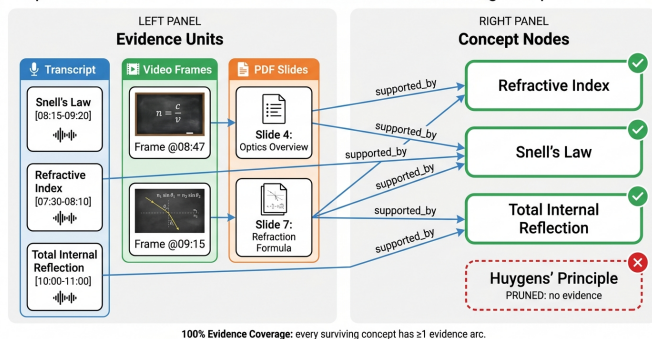


Fig. 3. Provenance enforcement on an optics lecture. Left: Evidence Units drawn from the transcript, video frames, and slides, each timestamped. Right: the concept nodes. A concept is kept only if at least one *supported_by* link reaches it from an Evidence Unit. Refractive index, Snell’s law, and total internal reflection are all attested and survive; “Huygens’ principle” has no supporting evidence because the lecture never taught it, so it is pruned. Every concept in the returned graph is therefore traceable to a specific moment in the source.

a large noisy one, and the ablation in Section III-C shows the trade is cheap: removing the rule costs only 0.02 Concept F_1 but lets many invented nodes back in.

The surviving edges may still form a cycle when distant segments give conflicting evidence. We break cycles with Tarjan’s algorithm, removing the lowest-scoring edge inside each strongly connected component until the graph is acyclic, which yields the final prerequisite directed acyclic graph.

III. EXPERIMENTS AND RESULTS

A. Dataset and Baselines

a) *Dataset.* We evaluate on 120 Khan Academy lectures across four STEM subjects: physics (31), biology (29), chemistry (32), and mathematics (28). They were chosen to span slide lectures, chalkboard derivations, and mixed formats, and they vary widely in length and concept density. Two subject-expert annotators watched each video in full and marked the named concepts and the directed prerequisite links between them; a third expert adjudicated disagreements, giving a gold standard of 1,847 concept-prerequisite pairs. Agreement before adjudication was Cohen’s $\kappa = 0.74$, in the substantial range and consistent with how subjective prerequisite labelling is. We split the corpus into 84 training and 36 test videos, tuned every hyperparameter on the training split alone, and used the test set only for the final numbers. Taking one annotator’s labels as gold and scoring the other on the unadjudicated test set gives a human ceiling of 0.831 F_1 on concepts and 0.619 on edges.

We compare ten systems. HTG is a deterministic, GPU-free floor that extracts noun chunks with spaCy, infers edges from cue phrases, and merges names by fuzzy matching. Three *LLM-Transcript* baselines (Gemma-4-E4B-it, Qwen3.5-9B, GLM-4.6V-Flash) prompt the model once on the full transcript with no visual input, no name merging, and no EWPI; this is current common practice. Three *LLM-Transcript+Captions*

baselines add captions of uniformly sampled frames, testing whether simply pouring visual descriptions into the prompt closes the gap. The remaining three are our full pipeline and two ablations of it: *No-MM* strips every visual signal, and *No-EWPI* accepts the language model’s raw prerequisite pairs without validation. The LLM baselines and our pipeline use the same model and concept taxonomy, so the gaps reflect architecture rather than prompt tuning or model choice.

We report three primary metrics. Concept F_1 (Soft) is micro-averaged F_1 over the test videos, counting a prediction correct if it matches a gold concept exactly or has cosine similarity at least 0.80 to one; Concept F_1 (Strict) counts only exact matches. Edge F_1 is directed: both endpoints must match gold concepts and the arrow must point the gold way. Two secondary checks accompany them: evidence coverage, the fraction of predicted concepts with at least one supporting Evidence Unit, which is our direct measure of hallucination suppression; and DAG validity, whether the returned graph is acyclic. Significance uses the Wilcoxon signed-rank test on per-video metric vectors (Shapiro–Wilk rejected normality for most systems), Bonferroni-corrected to $\alpha = 0.00625$. Seeds are fixed; mean processing is about 4.2 minutes and 1,200 tokens per video. Code, annotation guidelines, prompts, and splits are available.¹

B. Main Results

Table III gives the comparison, and Fig. 4 plots it. Four results stand out:

- *Near the human ceiling.* The full pipeline reaches Concept F_1 of 0.763 (soft) and 0.681 (strict) and Edge F_1 of 0.512, which is 91.8% of the human concept ceiling and 82.7% of the human edge ceiling.
- *A 16-fold edge gap.* The best baseline, GLM-4.6V-Flash on transcripts, reaches Edge F_1 of only 0.031. Our pipeline is 16 times higher on the metric that matters most.
- *Not a model-size problem.* All three baselines, from 4B to 9B parameters, land between 0.022 and 0.031 on edges, so the gap comes from how the prerequisite decision is made, not from model scale.
- *Naive captions hurt.* Adding captions of uniformly sampled frames makes every baseline slightly worse, because those captions mostly describe talking-head shots and static titles and dilute the prompt. The pipeline avoids this by choosing frames on content change and tying each caption to an Evidence Unit before any edge is inferred.

Every concept the pipeline returns links to a timestamped Evidence Unit, so evidence coverage is 100%.

C. Ablation and Statistical Validation

Table IV removes one component at a time. Every component helps, but unevenly:

- *EWPI recovers the edges.* Accepting the model’s raw prerequisite pairs (No EWPI) drops Edge F_1 from 0.512

¹<https://anonymous.4open.science/r/mywork-F50E>

TABLE III

SYSTEM COMPARISON ON THE 120-VIDEO BENCHMARK, WITH THE HUMAN INTER-ANNOTATOR CEILING FOR REFERENCE. MICRO-AVERAGED $F_1 \pm 95\%$ CI. BEST IN **BOLD**, SECOND BEST UNDERLINED. COST IS THE FRACTION OF SEGMENTS SENT TO THE LANGUAGE MODEL.

System	Model	Soft F_1	Strict F_1	Edge F_1	Cov.	Cost
HTG		0.224 $\pm$.018	0.101 $\pm$.014	0.014	N/A	0%
LLM Trans	Gemma 4 E4B it	0.298 $\pm$.021	0.119 $\pm$.016	0.022	N/A	100%
LLM Trans	Qwen3.5 9B	0.325 $\pm$.022	0.128 $\pm$.016	0.026	N/A	100%
LLM Trans	GLM 4.6V Flash	<u>0.341$\pm$.023</u>	<u>0.134$\pm$.017</u>	<u>0.031</u>	N/A	100%
LLM Trans+Cap	Gemma 4 E4B it	0.281 $\pm$.019	0.108 $\pm$.015	0.017	N/A	100%
LLM Trans+Cap	Qwen3.5 9B	0.302 $\pm$.020	0.114 $\pm$.015	0.019	N/A	100%
LLM Trans+Cap	GLM 4.6V Flash	0.318 $\pm$.021	0.121 $\pm$.016	0.024	N/A	100%
Full Pipeline	Gemma 4 E4B it + Qwen3 ASR 0.6B	0.763$\pm$.031	0.681$\pm$.028	0.512$\pm$.024	100%	45%
No MM	Gemma 4 E4B it + Qwen3 ASR 0.6B	0.651 $\pm$.028	0.572 $\pm$.025	0.301	100%	45%
No EWPI	Gemma 4 E4B it + Qwen3 ASR 0.6B	0.638 $\pm$.027	0.559 $\pm$.024	0.187	100%	45%
Human IAA		0.831 $\pm$.019	0.779 $\pm$.017	0.619	N/A	N/A

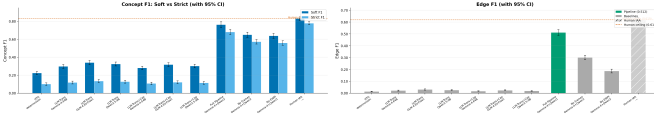


Fig. 4. Concept F_1 (left, soft and strict bars) and directed Edge F_1 (right) across the ten systems, with the human ceiling drawn as a dashed line (0.831 for concepts, 0.619 for edges). The full pipeline sits closest to both ceilings; the language-model baselines cluster near the bottom on edges, where the 16-fold gap is widest. Error bars are 95% confidence intervals.

TABLE IV

ABLATION (TOP) AND SIGNIFICANCE TESTS (BOTTOM). Δ EDGE IS THE CHANGE IN EDGE F_1 FROM REMOVING ONE COMPONENT. ALL LISTED COMPARISONS ARE SIGNIFICANT AT $p < 0.001$ (WILCOXON, BONFERRONI $\alpha = 0.00625$).

Ablation					
Config.	Soft	Strict	Edge	Cost	Δ Edge
Full pipeline	0.763	0.681	0.512	45%	—
No EWPI	0.638	0.559	0.187	45%	-0.325
No prov.	0.743	0.666	0.471	45%	-0.041
No routing	0.763	0.681	0.512	100%	0.000
No MM	0.651	0.572	0.301	45%	-0.211
No ER	0.420	0.388	0.445	45%	-0.067
Significance (Cohen's d)					
Comparison	Metric	p	d		
Full vs. GLM	Concept	< .001	1.84		
Full vs. GLM	Edge	< .001	2.31		
No EWPI vs. Full	Edge	< .001	1.43		
No MM vs. Full	Edge	< .001	1.12		

to 0.187, a fall of 0.325, larger than any other removal and two-thirds of the whole gap to the best baseline. The candidates are not just noisy, they are systematically backwards, and the four-signal vote is what corrects the direction.

- *Name resolution recovers the concepts.* Without it (No ER), Concept F_1 falls from 0.763 to 0.420, an even larger drop than EWPI causes on that metric, because each surface form of a concept becomes its own disconnected node. Deduplication is load-bearing, not cosmetic.
- *Visual signals help edges most.* Removing them (No MM)

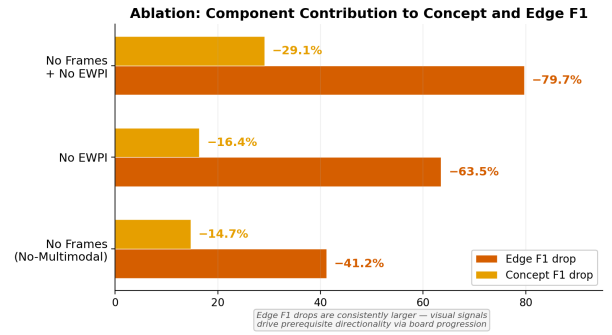


Fig. 5. Relative drop in Concept F_1 and Edge F_1 when each component is removed. The Edge F_1 bars are taller throughout, most sharply for EWPI and for the visual signals (No MM, 41% versus 15%), which is why we attribute prerequisite direction to multimodal evidence rather than to the language model.

costs 0.112 Concept F_1 but 0.211 Edge F_1 (15% versus 41% relative). With no board-write order, the visual signal V is gone and direction must rest on the temporal and discourse signals alone, which are weak on derivations.

- *Provenance is nearly free.* Dropping it (No Prov.) costs only 0.020 Concept F_1 while it prunes the invented nodes that would otherwise become spurious edge endpoints.
- *Routing is a pure cost lever.* No-Routing matches the full pipeline exactly on both metrics but at 100% language-model use instead of 45%, confirming that the band classifier escalates the right segments.

All comparisons in the table reach $p < 0.001$ with Cohen's $d \geq 1.12$, well above the $d = 0.8$ convention for a large effect.

D. Why the Language-Model Baselines Fail on Direction

The 16-fold edge gap deserves a concrete explanation rather than “the pipeline is better.” The root cause is that a bare transcript does not separate things mentioned together from things that depend on each other. The model sees that A was discussed before B and, reasonably for its training data, calls A a prerequisite of B . In a lecture that rule misfires often: instructors forward-reference a concept before its prerequisites are in place, return to foundational ideas to reinforce them, and

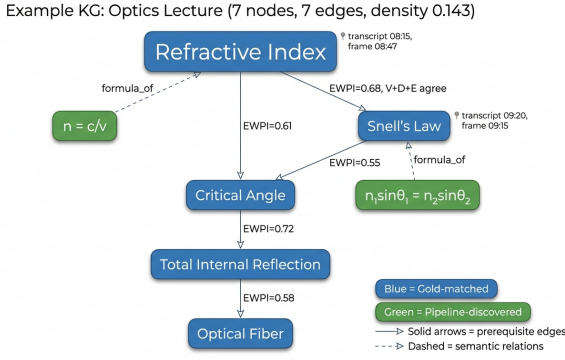


Fig. 6. Output graph for an optics lecture. Solid arrows are prerequisite edges, each labelled with the EWPI score that admitted it (for example, refractive index $\rightarrow$ Snell’s law at 0.68); dashed arrows are non-prerequisite semantic relations such as the formulas $n = c/v$ and $n_1 \sin \theta_1 = n_2 \sin \theta_2$. Blue nodes matched a gold annotation; green nodes were found by the pipeline. Each concept is annotated with the transcript time and frame it was grounded in, so the whole graph can be traced back to the lecture.

open with a motivating example before the definition it rests on. The No-EWPI number makes this quantitative, an Edge F_1 of 0.187 that matches a roughly 75% wrong-direction rate on admitted edges. Frame captions do not rescue the baselines either: they carry some sequence information, but without selective frames and an explicit board-progression measure it is too diffuse to fix the direction, as the Transcript+Captions results show.

IV. DISCUSSION AND CONCLUSION

Figure 6 shows what the pipeline produces for one optics lecture, and it makes the headline results concrete. The prerequisite chain runs refractive index $\rightarrow$ Snell’s law $\rightarrow$ critical angle $\rightarrow$ total internal reflection, each edge carrying the EWPI score that admitted it, and each concept tagged with the timestamp and frame it came from. Two formulas the lecture wrote on the board, $n = c/v$ and $n_1 \sin \theta_1 = n_2 \sin \theta_2$, attach to their concepts as semantic relations rather than prerequisites, and “Huygens’ principle” does not appear because the lecture never taught it.

Read together, the experiments point in one direction: getting prerequisites right depends far more on grounding the decision in evidence than on the size of the model. EWPI is the component that recovers edge direction, and its drop of 0.325 in Edge F_1 when removed is larger than the spread across any pair of baselines; its weights, fit on the training split, hold up across all four subjects, which suggests the signals it reads are not subject-specific. Provenance enforcement suppresses invented concepts for almost no cost in concept quality, name resolution turns out to be load-bearing rather than cosmetic, and routing buys roughly half the inference cost back at no measurable loss.

The study has clear limits. The benchmark is 120 Khan Academy lectures in four STEM subjects, all in English, so we cannot speak to other platforms, to humanities or professional

subjects where prerequisite structure may work differently, or to multilingual lectures. The gold standard comes from two annotators, and although $\kappa = 0.74$ is substantial, prerequisite labelling is partly a matter of pedagogical judgment. The EWPI weights were tuned on the training split and we have not measured how sensitive they are across subjects. The natural next steps are a wider and more varied benchmark, OCR of board text to strengthen the visual signal, and use inside a tutoring system, where a graph whose every edge is traceable to a moment in the lecture lets the system justify the order in which it teaches. The grounded graph is also a natural index for graph-aware retrieval [40], [30], [31] and a substrate for graph neural reasoning over concept dependencies [32], [33].

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